

ORIGINAL ARTICLE

Contrast-enhanced, single-shot LED array microscopy based on Fourier ptychographic algorithm and deep learning

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Abstract

LED array microscopes have the advantages of miniaturisation and low cost. It has been demonstrated that LED array microscopes outperform Köhler illumination microscopes in some applications. A LED array allows for a large numerical aperture of illumination. The larger numerical aperture of illumination brings the higher spatial resolution, but the lower image contrast as well. Therefore, there is a tradeoff between resolution and contrast for LED array microscopes. The Fourier ptychographic algorithm can overcome this tradeoff by increasing image contrast without sacrificing spatial resolution. However, the Fourier ptychographic algorithm requires acquisition of multiple images, which is time-consuming and results in live sample imaging challenging. To solve this problem, we develop contrast-enhanced, single-shot LED array microscopy based on the Fourier ptychographic algorithm and deep learning. The sample to be imaged is under illumination by all LEDs of the array simultaneously. The image captured is fed to several trained convolutional neural networks to generate the same number of images that are required by the Fourier ptychographic algorithm. We experimentally present that the image contrast of the final reconstruction is remarkably improved in comparison with the image captured. The proposed method can also produce chromatic-aberration-free results, even when an objective without aberration correction is used. We believe the method might provide live sample imaging with a low-cost approach.

KEYWORDS

contrast-enhanced, deep learning, Fourier ptychographic algorithm, LED array microscope

1 | INTRODUCTION

LED array illumination microscopes offer flexible applications that are superior to Köhler illumination microscopes in certain fields,^{1–8} such as cell culture monitoring,⁹ due to their miniaturisation and affordability advantages.^{10–12} However, LED array microscopes have a tradeoff between spatial resolution and image contrast, similar to the

Köhler-illumination-based bright-field microscope. Spatial resolution and image contrast are two of the most important parameters of a microscope. We note that the high-frequency components of the image within the resolution limit of a microscope are only useful if they have sufficient contrast. Commercial Köhler illumination microscopes usually recommend stopping down the condenser aperture to about 70% of the maximum aperture

diameter.^{13,14} The LED array microscope can also adjust the aperture size by controlling the number of 'ON'-state LEDs. As the image obtained is the intensity superposition of the images for a single LED illumination, multiple-LED illumination improves spatial resolution. However, the zero-frequency component of the resulting image is also superimposed, which leads to a decrease in image contrast. With the help of the Fourier ptychographic algorithm,¹⁵ both the spatial resolution and the image contrast can be enhanced.

However, Fourier ptychographic algorithm requires a series of images acquired by turning on a single LED in turn. This process is time-consuming and limits the application of live sample imaging. It has been demonstrated that multiple intensity distributions captured by the Fourier ptychographic microscopy (FPM) can be collected in a single shot by placing the lens array at the Fourier plane of the objective lens,¹⁶ or by using an objective lens array and a camera array.¹⁷ Nevertheless, the cost of these methods significantly increases. We note that full-colour FPM is also challenging for live sample imaging, because refocusing might be needed when the illumination wavelength changes particularly when the utilised object is not chromatic-aberration free.

Recently, researchers have aimed to use a small number of input images to obtain high-quality results directly through deep learning. Convolutional neural networks (CNNs) have been used for superresolution of microscopic images,^{18–22} replacing time-consuming reconstruction algorithms,^{23,24} and optimising parameters and illumination patterns of FPM,²⁵ among others. For example, a U-Net can be trained to directly reconstruct superresolution images from structured illumination microscopy (SIM) raw data using a small number of raw images.²⁶ It can reconstruct the isotropic differential phase contrast image by feeding an anisotropic differential phase contrast image through the U-Net.²⁷ Generative adversarial networks (GANs) can be trained to generate other structured illumination images by feeding a single structured illumination image.²⁸ However, most of these end-to-end deep learning approaches require a large-scale network model and dataset. The network model might be even complex when the imaging system has aberrations.

In this article, we present a contrast-enhanced, single-shot LED array microscopy based on the Fourier ptychographic algorithm and deep learning. Compared with the existing deep learning-based methods, our proposed method employs a simpler network, requires fewer training data and less computational time for network training. Furthermore, the use of a physical model allows for reliable reconstruction. Our method only requires a single shot of image by simultaneous illumination of multiple LEDs. Through the trained network, the images corre-

sponding to each single LED are generated from the single image captured. By feeding the generated images to the Fourier ptychographic algorithm, a contrast-enhanced image is reconstructed. To avoid repeated focusing, we also generate the corresponding defocused images for defocused wavelengths and correct defocused aberrations in the reconstruction process.

2 | METHODS

2.1 | Imaging system

Figure 1A shows a schematic diagram of the proposed microscopy. The microscope equips with a programmable LED array (9 mm spacing between adjacent LEDs), an objective lens (Nikon CFI Plan Fluor 4×, NA 0.13), a tube lens (Thorlabs, TLL200), and a monochromatic CMOS (Complementary Metal Oxide Semiconductor) camera (Point grey, FL3-U3-32S2M-CS, pixel size 2.5 μm). In Figure 1B, NA_{obj} denotes the numerical aperture of the objective lens and NA_{sys} denotes the synthetic numerical aperture of the system. We note that a synthetic aperture imaging can expand the numerical aperture of the microscope. To efficiently sample the energy distribution of natural images in the Fourier domain, we use an efficient rhombus sampling strategy in our system. The overlap rate is 71.89%, which meets the spectral overlap requirement for a successful reconstruction.¹⁵ Our system adopts a low-cost camera and a 13-element LED array to acquire bright-field images. We note that the camera is not sensitive to acquire dark-field images and, therefore, only bright-field images are acquired. We capture only one image with all 13 LEDs turned on simultaneously. We then use 13 trained artificial neural networks to generate the images, each of which corresponds to 1 of the 13 LEDs illumination, to conduct synthetic aperture imaging via the Fourier ptychographic algorithm. As Figure 1C shows, when the objective lens has chromatic aberration, repeated focusing is required for full-colour FPM. In our experiment, we find the focus position when the sample is under red LEDs illumination. The defocus distances for blue and green illumination are 5 and 15 μm , separately. The proposed method allows us to directly feed the defocused images to the networks and the undesired aberration can be corrected by the Fourier ptychographic algorithm.

2.2 | Network and loss function

U-Nets are commonly used for medical image segmentation,²⁹ because they are simple, effective, straightforward and computationally inexpensive com-

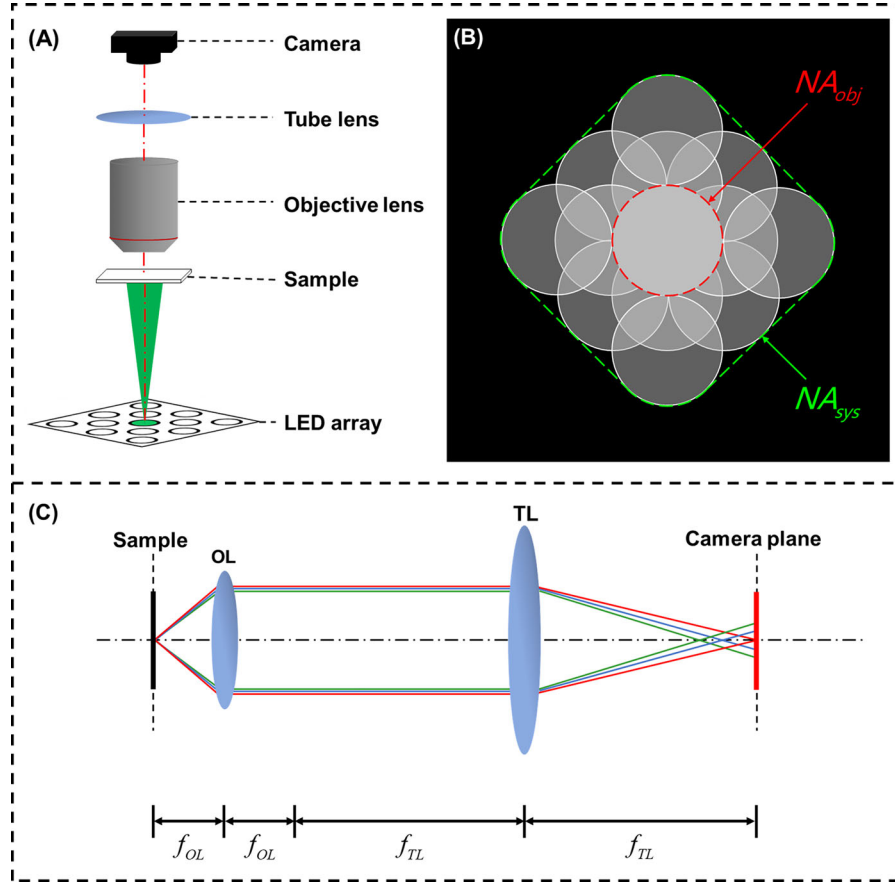


FIGURE 1 (A) The programmable LED array microscope, (B) synthetic aperture schematic, (C) chromatic-aberration schematic.

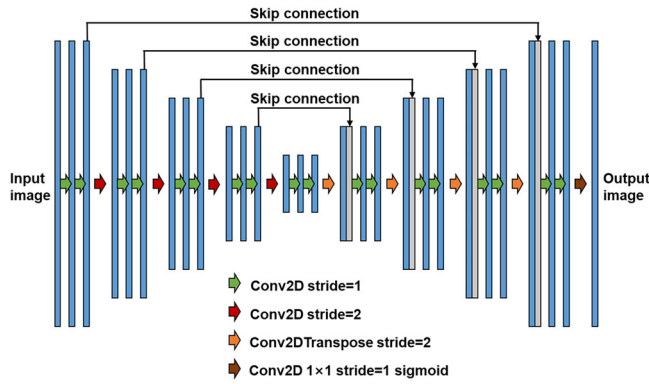


FIGURE 2 Architecture of the U-Net.

pared with traditional convolutional neural network models. We use 13 U-Nets to generate 13 images, each of which corresponds to single LED illumination, from the captured image. The U-Nets share the same architecture shown in Figure 2. We note that we replace the activation function of the last layer with a sigmoid function. We combine the mean absolute error (MAE) and the structural similarity index (SSIM) as our loss function. Additionally, we use the MAE of the Fourier spectrum as

the loss function to avoid raster artefacts in the generated images and improve the high-frequency details of the output images. Specifically, the loss function is defined as

$$l = \lambda_1 l_{MAE} + \lambda_2 l_{FMAE} + \lambda_3 l_{SSIM}, \quad (1)$$

where

$$l_{MAE} = \frac{\sum_{i=1}^W \sum_{j=1}^H |X_{ij} - Y_{ij}|}{W \times H}, \quad (2)$$

$$l_{FMAE} = \frac{\sum_{u=1}^W \sum_{v=1}^H |F\{X\}_{uv} - F\{Y\}_{uv}|}{W \times H}, \quad (3)$$

$$l_{SSIM} = 1 - \frac{(2\mu_x\mu_y + c_1)(2\sigma_{xy} + c_2)}{(\mu_x^2 + \mu_y^2 + c_1)(\sigma_x^2 + \sigma_y^2 + c_2)}, \quad (4)$$

where λ_1 , λ_2 and λ_3 are the hyperparameters that control the relative weights of each loss components; W and H denote the width and height of the image, respectively; X_{ij} and Y_{ij} represent the greyscale values of the ground truth image X and the generated image Y at point (i, j) ; F denotes the 2D Fourier transform; $F\{X\}_{uv}$ and $F\{Y\}_{uv}$ represent the absolute intensity value at Fourier coordinates

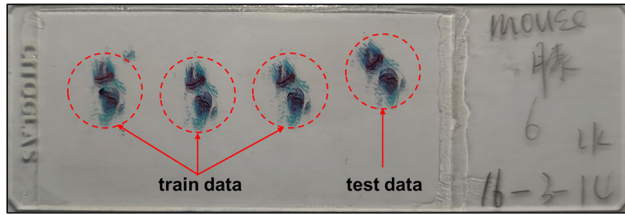


FIGURE 3 Slide of mouse knee samples. The first three regions are used for training and the rest region is used for testing.

(u, v) ; μ and σ^2 are the mean and the variance of x and y , respectively; σ_{xy} is the covariance of x and y and c_1 and c_2 are the regularisation parameters.

2.3 | Data preparation

We train two sets of U-Net for resolution test chart imaging and biological sample imaging, respectively. To train the U-Nets for resolution test chart imaging, we randomly capture 500 images of a USAF-1951 resolution test chart, among which 400 images are used for network training and 100 for network testing. To train the U-Nets for biological sample imaging, we use a slide which consists of 4 samples, as Figure 3 shows. We randomly collect 225 images of each sample. The networks are trained for red, green and blue channel reconstruction separately. For each chromatic channel, we use a total of 675 images of the first three (from left to right) samples for training and 225 images of the last sample for testing. All images are 8-bit greyscale and 512×512 pixels in size. For each network, the input is the image captured when all 13 LEDs are turned on, and the label is the image captured when a single LED is turned on. When all LEDs are illuminated simultaneously, the camera exposure time is set to $5000 \mu\text{s}$, and the gain is set to 0 dB. When a single LED is turned on, the exposure time is set to $19,988 \mu\text{s}$, and the gain is set to 10 dB.

2.4 | Training and reconstruction

The U-Nets are trained on a desktop computer (CPU: Intel® Core™ i5-11260H, GPU: NVIDIA RTX3050 with 8GB RAM) using Tensorflow-GPU 2.4.0 based on Python 3.7.13. Each U-Net is trained for 10 epochs, which takes ~ 9 min. Figure 4 illustrates how single-shot synthetic aperture imaging is implemented using the U-Net.

3 | RESULTS

As shown in Figures 5 and 6, the size of the reconstructed images is 2048×2048 . For a fair comparison, we upsample the images captured to the same size via the bicubic interpolation algorithm.³⁰ As the results show, when a single LED is on, the finest distinguishable structure is Group 8 Element 1 while it is Group 8 Element 4 when all LEDs are on. On the contrast, Group 8 Element 5 can be well resolved in the image reconstructed by the proposed method. The results indicate that we can successfully recover the sample's high-resolution and contrast-enhanced image with a single shot.

To validate the proposed method in biological sample imaging, we retrained the 13 U-Nets using mouse knee sample images. As shown in Figure 7, the trained U-Nets can successfully generate images with fine details. For both focused (R) and defocused (G and B) channels, the images generated by the resulting networks are effective in reducing noise while retaining the details.

For traditional full-colour FPM experiments, a series of low-resolution images has to be captured using different wavelengths for illumination. This process can be time-consuming. However, our proposed method requires only one image to be captured for each illumination wavelength. Additionally, it is not necessary for the proposed method to adjust the focus when the illumination

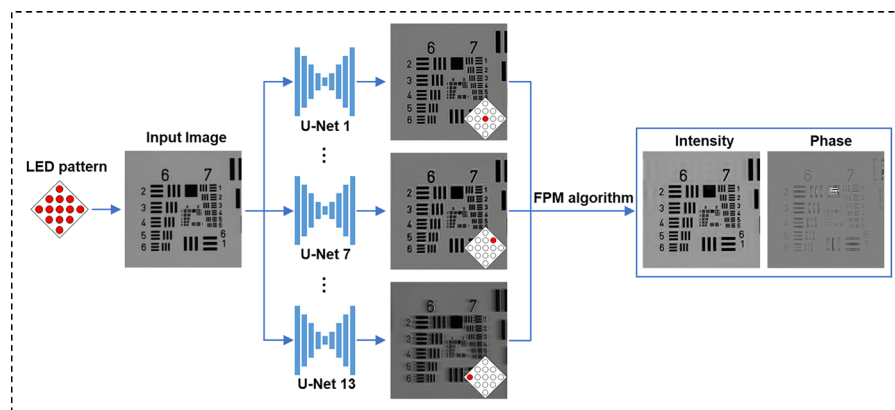


FIGURE 4 U-Net-based single-shot synthetic aperture imaging based on Fourier ptychographic algorithm.

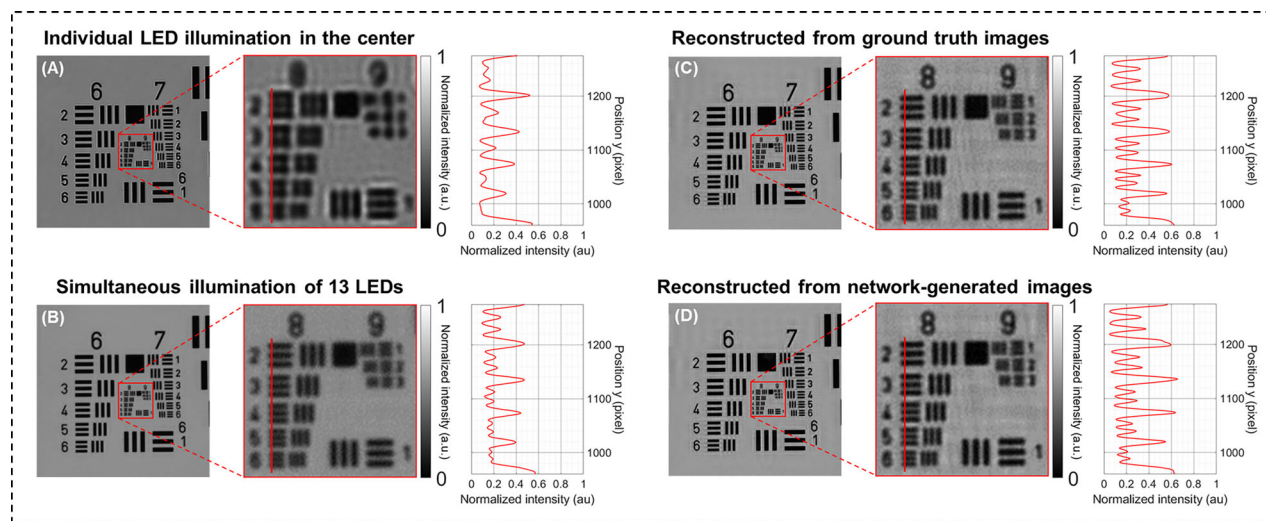


FIGURE 5 Resolution and contrast analysis of the reconstructed result of USAF-1951 resolution test target. (A) Individual LED illumination in the centre, (B) simultaneous illumination of 13 LEDs, (C) the intensity reconstructed by the captured 13 ground truth images, (D) the intensity reconstructed by the network-generated 13 images.

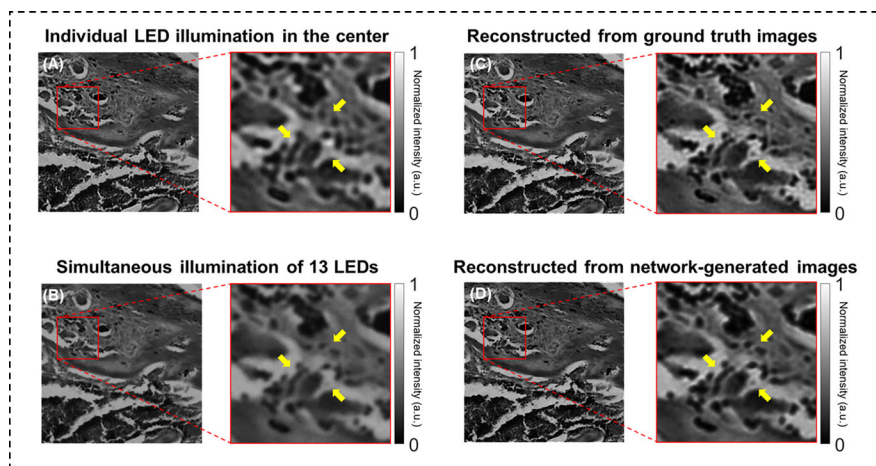


FIGURE 6 Contrast analysis of the result of the reconstruction of mouse knee (red wavelength focused and illuminated). (A) Individual LED illumination in the centre, (B) simultaneous illumination of 13 LEDs, (C) the intensity reconstructed by the captured 13 ground truth images, (D) the intensity reconstructed by the network-generated 13 images.

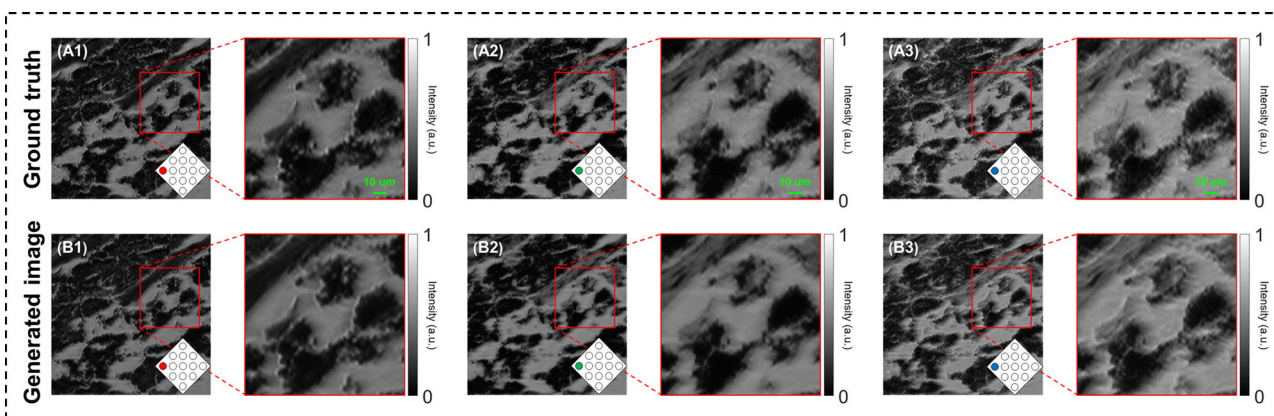


FIGURE 7 Images are generated by the network for the largest illumination angle (corresponding to the image with the most high-frequency components). The first row (A1)—(A3) are the ground truth images and (B1)—(B3) are the corresponding network-generated images.

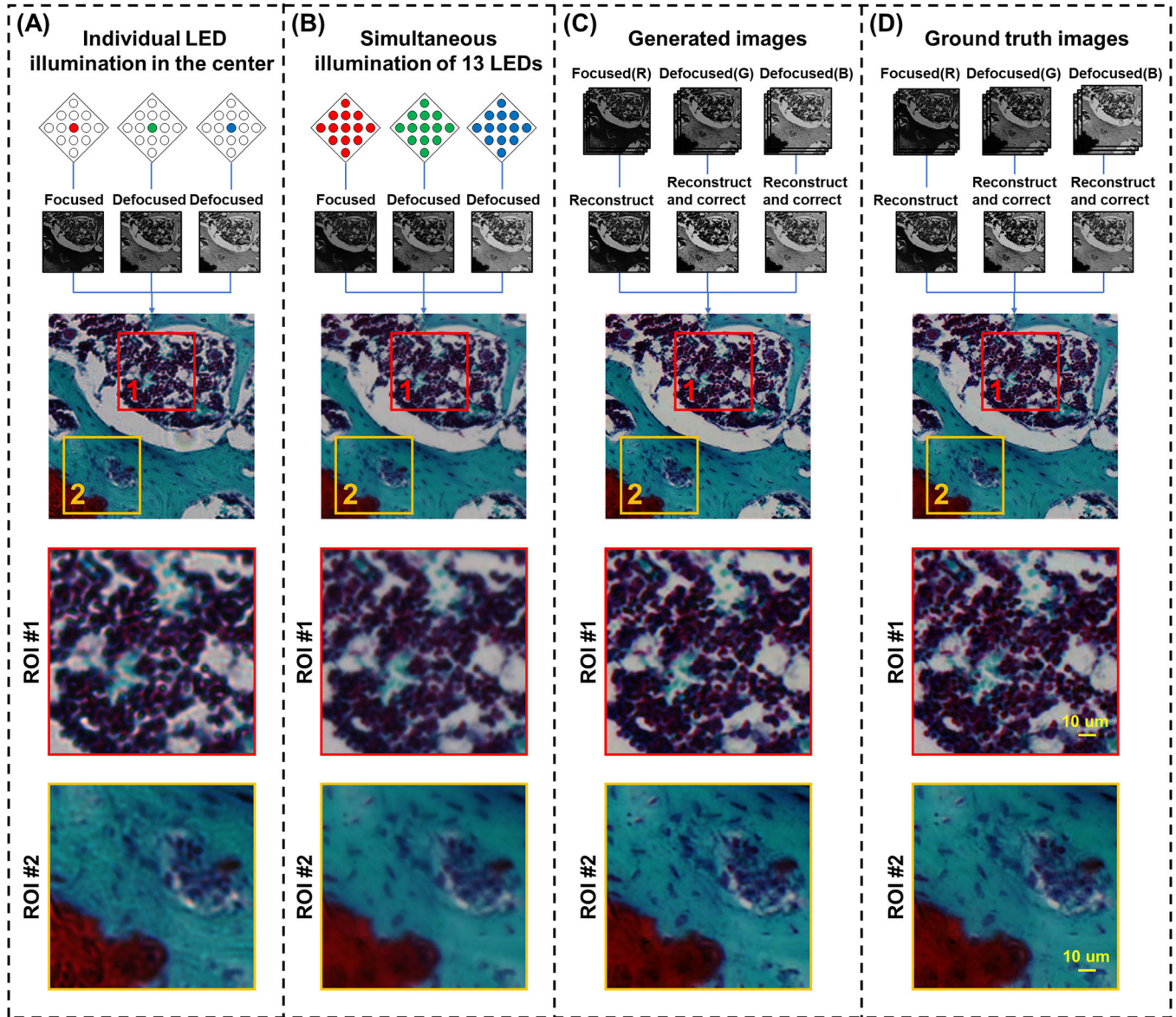


FIGURE 8 Comparison of Fourier ptychographic algorithm-based LED array colour microscopy results of mouse knee. (A) Individual LED illumination in the centre, (B) simultaneous illumination of 13 LEDs, (C) reconstruction from the generated images and (D) reconstruction from the ground truth 13 images.

wavelength changes, as the undesired chromatic aberration can be removed by the Fourier ptychographic algorithm. The basic idea is to digitally introduce a defocus phase factor to the pupil plane to model the connection between the spectral subregion of the reconstructed image and the intensity data with aberrations, thus enabling an aberration-free reconstructed image.¹⁵ As Figure 8 shows, the results by the proposed method are close to those by the traditional FPM.

We use SSIM to quantitatively evaluate the quality of the network-generated images. The images captured by a single LED illumination are used as the reference. The results are shown in Table 1. Moreover, we quantitatively evaluate the image contrast by using enhancement measure evalu-

ation (EME).³¹ EME measures the average contrast in an image by dividing the image into subblocks. The image X is divided into $M \times N$ subblocks x_{ij} with the same fixed size. Then we calculate the ratio of the maximum to the minimum greyscale value of each subblock. The higher the EME, the better the overall contrast of the image. The EME is defined as

$$EME(X) = \frac{1}{M \times N} \sum_{i=1}^M \sum_{j=1}^N 20 \ln \frac{\max(x_{ij})}{\min(x_{ij} + \delta)}, \quad (5)$$

where $\max(x_{ij})$ and $\min(x_{ij})$ are the maximum and minimum values of pixel intensity in subblock x_{ij} , respectively. δ is set to 0.0001 to avoid the division-by-zero exception.

TABLE 1 SSIM and MAE between the generated images and the ground truth images.

Image channel	Average SSIM	Standard deviation of SSIM for 13 groups of generated images	MAE
R (generated)	0.88	0.01	1.89
G (generated)	0.87	0.03	1.96
B (generated)	0.77	0.05	3.43
R (reconstructed)	0.89	N.A.	2.96
G (reconstructed)	0.93	N.A.	2.37
B (reconstructed)	0.91	N.A.	2.79

TABLE 2 EME comparison of different imaging methods for the mouse knee.

Captured (13 LEDs ON)	Reconstructed from 13 generated images	Reconstructed from 13 captured images
4.92	8.10	8.33

According to the reference,³¹ the size of each subblock is set to 8×8.

We conduct FPM reconstruction for each chromatic channel using the network-generated images. 225 full-colour images of the mouse knee are obtained. As shown in Table 2, the quality of the reconstruction by the generated images is close to that by the images captured, which demonstrates the effectiveness of our method.

4 | DISCUSSION

We note that the involvement of dark-field images in traditional FPM reconstruction is the key to spatial resolution improvement. However, due to the insufficient sensitivity, the camera used in our experiment is not able to capture dark-field images at a high signal-to-noise ratio. As a result, we are currently unable to demonstrate the proposed method in resolution-enhanced imaging.

5 | CONCLUSION

In this article, we report an implementation of low-cost LED array microscopy. The method is assisted by deep learning to achieve single-shot FPM. As such, the achievable image contrast is enhanced and the data acquisition time is greatly reduced. In addition, the method can reconstruct aberration-free results with an objective without chromatic-aberration correction. We believe the method might provide live sample imaging with an effective low-cost approach.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

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